LM Physics of Climate

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7 Radiative heating

7.1 Radiative equilibrium equation

In Sections 3 and 4, we saw how convective energy transport shapes the temperature profile of an atmosphere. Here, we will determine how radiation shapes the temperature profile in the parts of an atmosphere where radiative energy transport dominates. We will assume that the atmosphere is in radiative equilibrium, i.e. for each layer of the atmosphere the total energy absorbed is equal to the total energy emitted. In the absence of radiative equilibrium, the atmosphere heats up/cools down because it emits less/more energy than it receives.

When calculating radiative equilibrium, we need to consider the total energy emitted/absorbed across *all wavelengths* and *all directions*. Let's start by considering the energy emitted by the gas. For a given frequency, ν, we integrate the radiation emitted in all directions:

$$\begin{aligned}\text{radiation out} &= \int \kappa_\nu B_\nu \mathrm{d}\Omega \\ &= \int_0^{2\pi}\int_0^{\pi} \kappa_\nu B_\nu \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= 4\pi\kappa_\nu B_\nu, \end{aligned} \tag{7.1}$$

where we have used the fact that blackbody radiation is isotropic.

Similarly, the incoming radiation integrated over all directions is:

$$\begin{aligned}\text{radiation in} &= \int \kappa_\nu I_\nu \mathrm{d}\Omega \\ &= \int_0^{2\pi}\int_0^{\pi} \kappa_\nu I_\nu \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= 4\pi\kappa_\nu J_\nu, \end{aligned} \tag{7.2}$$

where in the last step we define

$$J_\nu = \frac{1}{2}\int_{-1}^{1} I_\nu \mathrm{d}\mu_\theta, \tag{7.3}$$

also known as the *mean intensity*.

For a layer of the atmosphere to be in radiative equilibrium, we therefore require the incoming and outgoing energy (integrated over all wavelengths) to balance, $\int_0^\infty (\text{in} - \text{out})\mathrm{d}\nu = 0$. Using Equations 7.1 and 7.2 we get:

$$\int_0^\infty \kappa_\nu (J_\nu - B_\nu)\mathrm{d}\nu = 0. \tag{7.4}$$

This is known as the radiative equilibrium equation. Note that it only includes fluxes from *radiative* energy transport. A more generalised form of this equation is *radiative-convective equilibrium*, in which convective fluxes are also added to the balance.

7.2 Heating rate

When a gas is not (yet) in radiative equilibrium, it heats up or cools down. We can calculate the heating/cooling rate by finding the difference between the incoming and outgoing fluxes for a given layer of gas in the atmosphere. To do this, we will return to the two-stream equations from Section 6.

The two-stream equations (Equations 6.9 and 6.10) can be solved using an integrating factor of $e^{\overline{\tau}}$ for the upward flux equation and $e^{-\overline{\tau}}$ for the downward flux equation, respectively (similarly to how we got to Equation 6.1). At the boundaries, we set $\overline{\tau} = 0$ at the base of the atmosphere, and $\overline{\tau} = \overline{\tau}_\infty$ at $z = \infty$ (or $P = 0$). This results in the following solutions for $\mathcal{F}_+$ and $\mathcal{F}_-$:

$$\mathcal{F}_+(\overline{\tau}, \nu) = \mathcal{F}_+(0)e^{-\overline{\tau}} + \int_0^{\overline{\tau}} \pi B_\nu(T(\overline{\tau}'))e^{-(\overline{\tau}-\overline{\tau}')}\mathrm{d}\overline{\tau}' \tag{7.5}$$

$$\mathcal{F}_-(\overline{\tau}, \nu) = \mathcal{F}_-(\overline{\tau}_\infty)e^{-(\overline{\tau}_\infty-\overline{\tau})} + \int_{\overline{\tau}}^{\overline{\tau}_\infty} \pi B_\nu(T(\overline{\tau}'))e^{-(\overline{\tau}'-\overline{\tau})}\mathrm{d}\overline{\tau}'. \tag{7.6}$$

Next, we define the *heating rate*, i.e. the amount of energy added or removed from an atmospheric layer per unit time. The heating rate per unit optical thickness per unit frequency is

$$h_\nu = -\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(\mathcal{F}_+ - \mathcal{F}_-). \tag{7.7}$$

It is also useful to express the heating rate per unit mass instead of per unit optical depth:

$$\begin{aligned} H_\nu &= -\frac{1}{\rho}\frac{\mathrm{d}}{\mathrm{d}z}(\mathcal{F}_+ - \mathcal{F}_-) \\ &= g\frac{\mathrm{d}}{\mathrm{d}P}(\mathcal{F}_+ - \mathcal{F}_-) \\ &= g\frac{\mathrm{d}\overline{\tau}}{\mathrm{d}P}\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(\mathcal{F}_+ - \mathcal{F}_-) \\ &= \frac{\kappa_\nu}{\overline{\mu}_\theta}h_\nu. \end{aligned} \tag{7.8}$$

Note that when a gas is in radiative equilibrium, the heating rate is zero.

7.3 The optically thin limit

Consider the solutions to the two-stream equations (Equations 7.5 and 7.6) in the limit of an optically thick gas, i.e. $\overline{\tau}_\infty \ll 1$. In this limit, the exponential terms in Equations 7.5 and 7.6 can be expanded as a Taylor series. We will be working to first order in $\overline{\tau}$, which means that the terms outside the integral need to be expanded up to $\mathcal{O}(\overline{\tau})$, and the terms inside the integral need only be expanded to $\mathcal{O}(1)$. The result is:

$$\mathcal{F}_+(\overline{\tau}, \nu) = \mathcal{F}_+(0)(1 - \overline{\tau}) + \int_0^{\overline{\tau}} \pi B_\nu(T(\overline{\tau}'))\mathrm{d}\overline{\tau}' \tag{7.9}$$

$$\mathcal{F}_-(\overline{\tau}, \nu) = \mathcal{F}_-(\overline{\tau}_\infty)(1 - (\overline{\tau}_\infty - \overline{\tau})) + \int_{\overline{\tau}}^{\overline{\tau}_\infty} \pi B_\nu(T(\overline{\tau}'))\mathrm{d}\overline{\tau}'. \tag{7.10}$$

Note that in Equation 7.9, the first term represents flux from the surface being slightly attenuated by the atmosphere, and the second term represents the thermal emission from the atmosphere. The equivalent is true for Equation 7.10.

We now define a useful quantity - the *mean emission* temperature, $\overline{T}_\nu$, which is defined such that

$$B_\nu(\overline{T}_\nu) = \frac{1}{\overline{\tau}_\infty}\int_0^{\overline{\tau}_\infty} B_\nu(T(\overline{\tau}'))\mathrm{d}\overline{\tau}'. \tag{7.11}$$

We can therefore write the flux which escapes at the top of the atmosphere, $\mathcal{F}_{+,\infty}$, and the flux which is radiated back towards the surface, $\mathcal{F}_{-,s}$, as:

$$\mathcal{F}_{+,\infty} = (1-\overline{\tau}_\infty)\mathcal{F}_{+,s} + \overline{\tau}_\infty \pi B_\nu(\overline{T}_\nu) \tag{7.12}$$

$$\mathcal{F}_{-,s} = (1-\overline{\tau}_\infty)\mathcal{F}_{-,\infty} + \overline{\tau}_\infty \pi B_\nu(\overline{T}_\nu). \tag{7.13}$$

From Equations 7.9 and 7.10, the heating rate of the atmosphere is

$$\begin{aligned} H_\nu &= -\frac{\kappa_\nu}{\overline{\mu}_\theta}\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(\mathcal{F}_+ - \mathcal{F}_-) \\ &= \frac{\kappa_\nu}{\overline{\mu}_\theta}(\mathcal{F}_+(0) + \mathcal{F}_-(\overline{\tau}_\infty)) - \frac{\kappa_\nu}{\overline{\mu}_\theta}\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}\left(\int_0^{\overline{\tau}} \pi B_\nu(T(\overline{\tau}'))\mathrm{d}\overline{\tau}' - \int_{\overline{\tau}}^{\overline{\tau}_\infty} \pi B_\nu(T(\overline{\tau}'))\mathrm{d}\overline{\tau}'\right) \\ &= \frac{\kappa_\nu}{\overline{\mu}_\theta}\left[(\mathcal{F}_{+,s} + \mathcal{F}_{-,\infty}) - 2\pi B_\nu(T(P))\right], \end{aligned} \tag{7.14}$$

where in the last step we have used the Leibniz rule for differentiating with respect to an integral limit: $\frac{\mathrm{d}}{\mathrm{d}x}\int_a^{h(x)} f(x')dx' = f(h(x))\cdot\frac{\mathrm{d}h}{\mathrm{d}x}$. For conciseness, we have also rewritten $\mathcal{F}_+(0) = \mathcal{F}_{+,s}$ and $\mathcal{F}_-(\overline{\tau}_\infty)) = \mathcal{F}_{-,\infty}$.

The positive terms in this equation represent the heating from the top and bottom of the atmosphere, while the B_ν term represents the radiative cooling term. Note that if $\mathcal{F}_{+,s} + \mathcal{F}_{-,\infty}$ is larger than $2\pi B_\nu(T(P))$, the gas will heat up ($H_\nu > 0$). Once the temperature has increased enough such that $2\pi B_\nu(T(P))$ balances the heating term, $H_\nu = 0$ and the gas returns to radiative equilibrium (unless it is disturbed by other processes).